

Q.1

a)

→ Process planning :- "process planning is the activity determining appropriate sequence and manufacturing operation to convert raw material into finished product is called as process planning".

b)

→ Hybrid CAPP :-

(Computer assisted process planning)

- Hybrid CAPP is the combination of variant (retrieval) approach and generative approach
- This hybrid CAPP balance between both practically.

c)

→ Meshing :- (Discretisation)

→ meshing is the process in which complex geometry convert into small part of element is called meshing.

d)

→

A = 3 ;

B = 7 ;

C = 2 ;

* $X = A + B * C$

→ output

$X = 17$

e)

→

mirror tool is observing the view of object by mirroring.

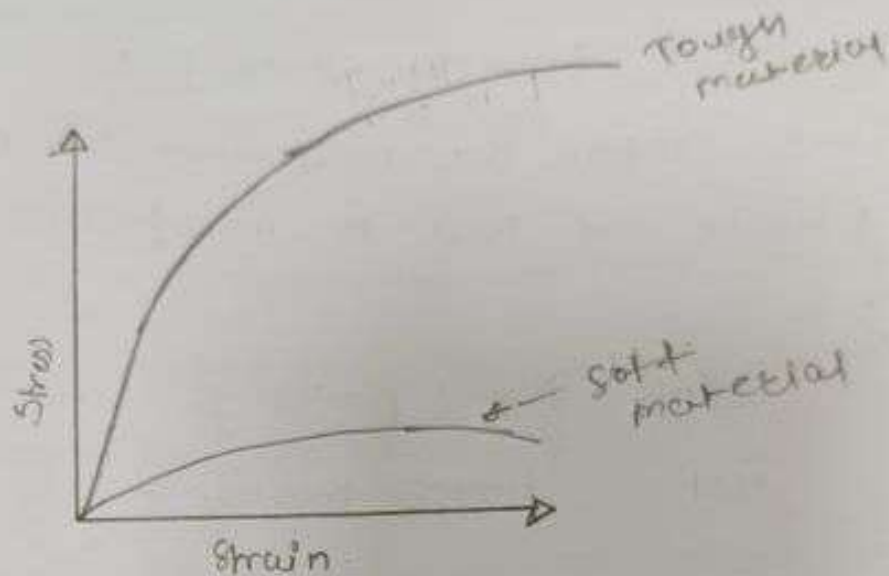
→ Requirement → to view the different reference plane of the object.

f)

→

clearvars is the command which is used to clear all variables from the matlab.

8)



b)



0.2 % offset yield method :-

"This method is used for determining of the yield strength of the material who does not have sharp or clear yield point on stress strain curve. is called as 0.2 % offset yield method."

9)



Toughness :- "Toughness is define as the in stress strain graph the region show absorb energy per unit volume, is called as Toughness."

$$\text{Toughness} = \frac{\text{Absorb energy}}{\text{volume}}$$

j)

→

G codes (Geometric codes) :-

That control the movement of machine
(Where and how to move tool)

eg:-

G00 - Rapid positioning

G01 - linear movement

M codes (Miscellaneous codes) :-

That code control the function of
machine (spindle on/off, coolant, program
stop)

M00 - Stop Program

M04 - Spindle On

M05 - Spindle off

k)

→

CNC = (Computer Numerical control) tech.

l)

→

Type of
element :-

1 D	2 D	3 D
line	- Triangle - Quadrilateral	- Tetrahedron - Hexahedron

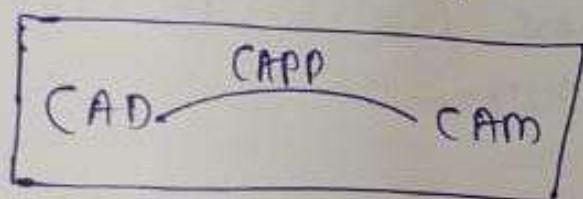
Q.3 //

→ CAPP (Computer assisted Process planning)

→ It is computer assisted technology used to automate or assist process planning is called CAPP.

→ it is the link between CAD and CAM.

→ it act as bridge between CAD & CAM.



Advantages of the CAPP :-

① Automation of the process planning :-

- CAPP reduce the manual work load
- also decrease the human dependency

② improve consistency :-

- Autogenerate process planning
- Reduce planner to planner variation.

3) Reduce lead time :-

- Rapid generation of process planning operation sequence that minimize the manufacturing time.

4) Optimization of cost :-

- selectively select the tool (machinability) that reduces the manufacturing cost.

5) Knowledge retention :-

- Standard and precise procedure flow into system.

6) flexibility :-

- Adapt any new changes in the process, operation, easily.

CAPP ~~Archite.~~

Architecture :-

- CAPP having various modules & layers that interact with CAD/CAM environment.

a) Input :-

- material specification
- production volume
- machinability tooling

b) Database :-

- material database
- process database
- part / feature

c) Knowledge Base :-

used decision logic, rules, Algorithm to select automatic sequence of operations

d) process planning :-

- Operation selection & operation
- set up process
- Time & cost estimation

e) Output-

- work sheet
-

CAPP is having following Two main Approaches.

① Variant (Retrival) approaches :-

- In this approach is based on group of technology.
- (i.e) part with similar design and manufacturing characteristics is group into families.
- Standard Process plans store for each family.
- for new part the existing plans are removed or modified.

② Advantages :-

- Use of existing knowledge
- easy to implement
- Used for batch Process.

Steps :-

- 1) classify each family by using code system
- 2) store the standard process planes to the each family
- 3) modify & retrieved as according.

⊕ limitations :-

- Not fully automatic
- Required initial classification system
- adaptability issue.

② Generative Approach :-

- Generative approach is knowledge & AI technique fully.
- In this approaches for generation of process planes may use AI.

- Automatic generate the process planes based on Algorithm, Decision logic, rules.
- This system analyse the Structure material, geometry, for generation of new planes from the scratch.

② Advantages :-

- It is fully automated
- Consistent
- used in computer integrated manufacturing.

③ Limitations :-

- 1) Required high description data.
- 2) High initial cost.

Q.4

→ CNC (Computer Numerical Control) tech.

→ it is the process given set of numerical instruction to the machine for (turning, cutting, milling etc).

⇒ CNC machine having two types of codes

① Generative (G codes) or

- It is control the machine movement
- How to move, where to move etc.

eg:

G00 - Rapid positioning

G01 - linear movement

G02 - circular interpolation (clockwise)

G03 - circular interpolation (anticlockwise)

G04 - Dwell (pause)

② Miscellaneous code (M-codes) :-

- This codes control the function of machine (spindle on/off, coolant on/off, ~~stop~~ end program).

eg :-

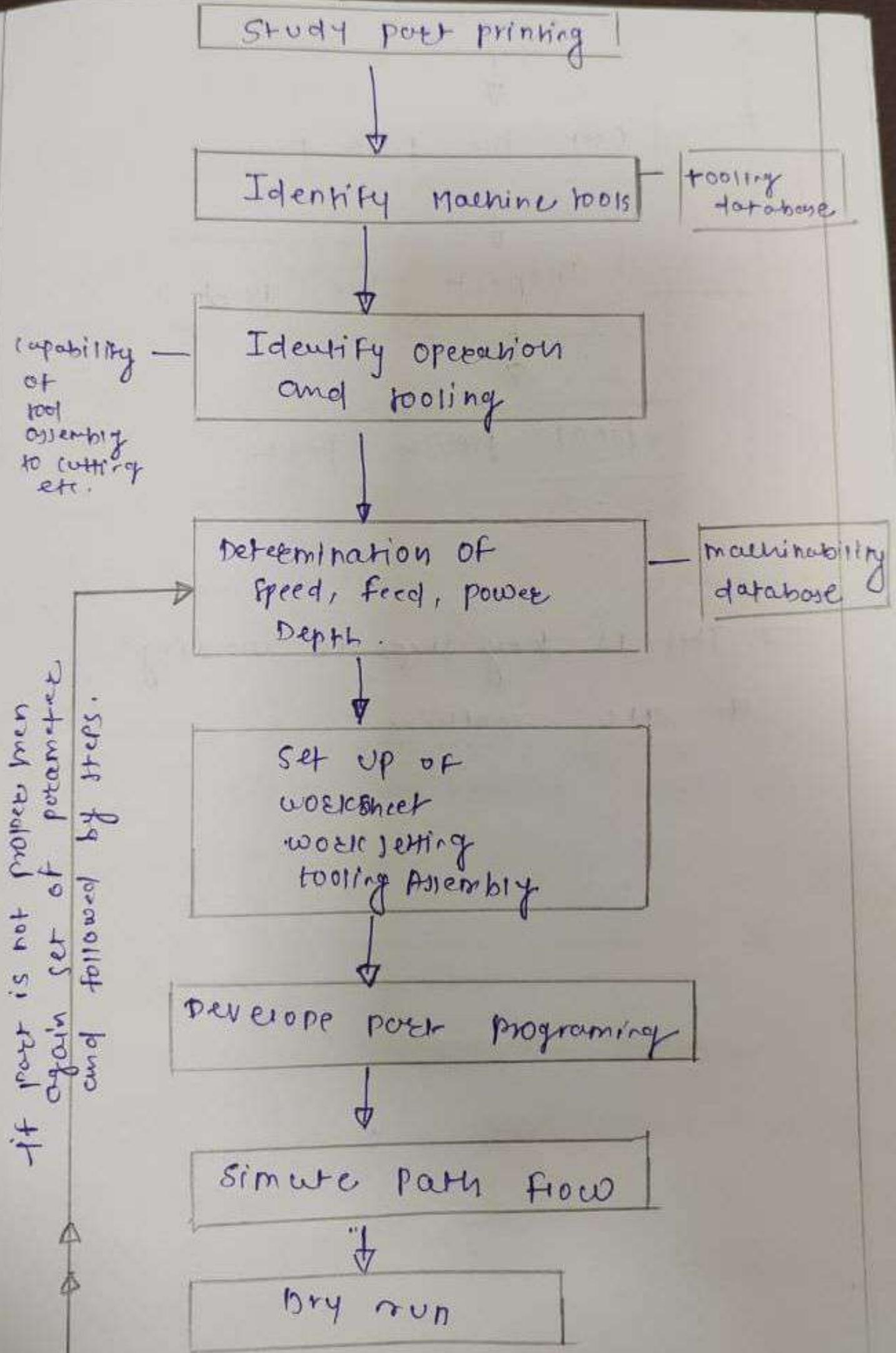
M00 - end program

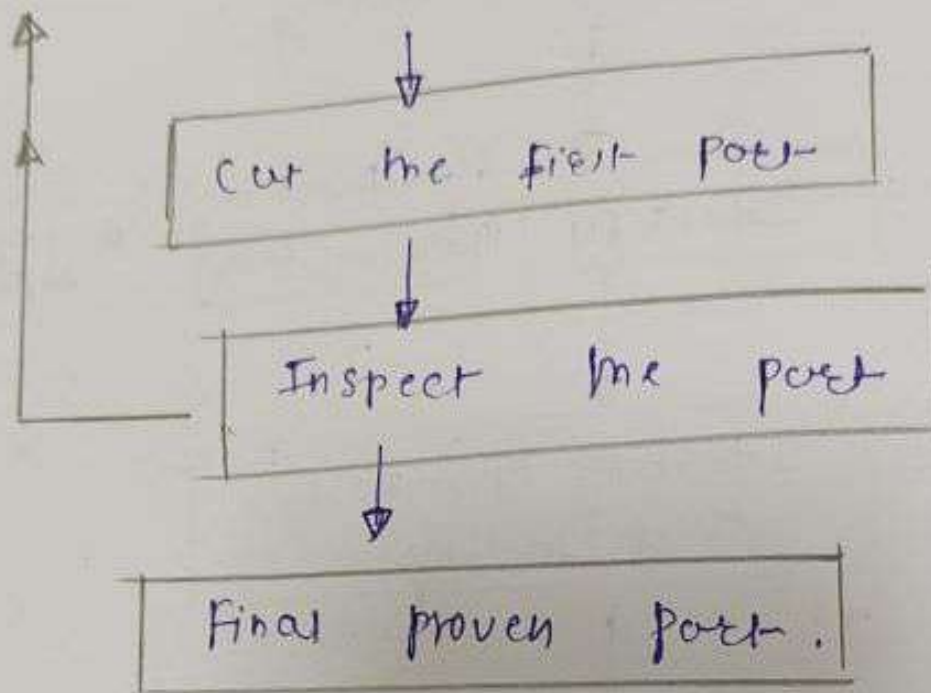
M03 - spindle on (clockwise)

M04 - spindle on (counter clockwise)

M05 - spindle off.

③ following are the key steps of workflow
CAD → CAM → CNC programming.





→ This is key steps in working of CNC machine.

Q.5)

→ Finite Element Analysis [FEA] :-

- Finite Element analysis is the computer numerical based method in which we estimate the structure and material behavior by external ^{world} force such as.
(Temperature, pressure, compression etc).
- Instead of solving complex equation for object we divide into small part called as element.
- Computer compute the each element behaviour and then combine to estimate whole system behaviour.

- following are the step by step process of (FEA) :-

① Geometry creation :-

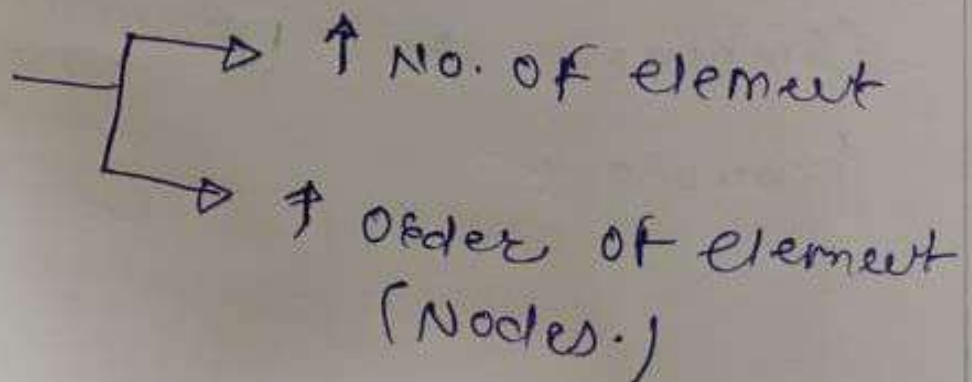
- Create a 3D model by using CAD software such as ProE, MATLAB, etc.

② Meshing :-

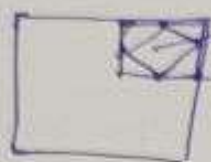
- Meshing is the process in which complex geometry convert into simple small part.

- In this process, complex geometry convert into number of nodes where calculation occurs.

- increase meshing



eg:



element
Nodes

③ Element Type :-

Element Type	Shape	Use / Application	Example
1D	lines.	Use for simply transfer of workload to frame, &	Stem construct, catheter wire
2D	- Triangle - Quadrilateral	used for thin layer like (membrane wall, shell etc)	Thin layer drug patches
3D	Tetrahedron Hexahedron	used for solid material with having volume	Bone Implant.

④ Application of material properties

→ Assign the material properties it
whether mechanically and
Thermally. by using (Poisson ratio,
thermal conductivity etc).

⑤ Boedec constrain

— In this detect how material
react to load.

⑥ Solving

— In this process computer etc
calculate behaviour of each element

⑦ Post processing

— In this process demonstrate the
the result of compression,
pressure etc.

Applications of FEA :-

- 1) Strain Analysis of Insuline pen, and microneedle.
- 2) Studying stress strain properties of tissue and Bone deformation due to implant
- 3) Stress & compression estimation of Compression tablet..

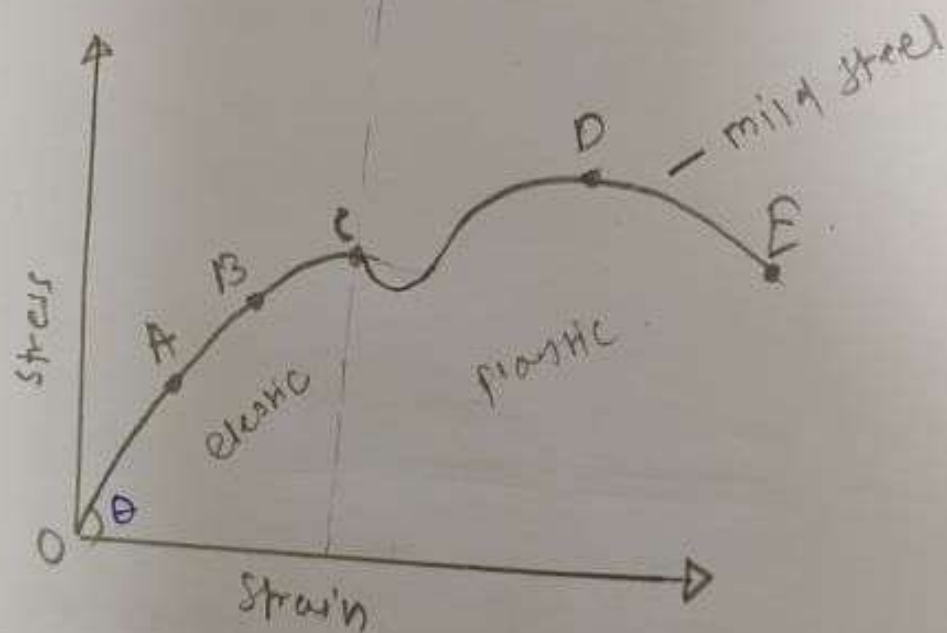
Q.6)



i) $\rightarrow$

Stress strain curve demonstrate that material structure subjected to the increase tensile strength, compression force etc.

- following are the stress strain diagram of mild steel.



$\rightarrow$ In this stress strain curve the (O) zero is the origine of Test, then next is

(A)

Point

(A)

- Proportionality limit :- In

that the stress strain curve is ~~direct~~ show that stress is directly proportional to the strain that follow hooke's law.

- In that material show perfect-elastic property.

Point

(B)

- Elasticity limit :- In that

that region the material having ability to return into its original shape after unloading.

- In this region material will show elastic behaviour.

Point (C) :- Yield point :- Yield

means maximum stress apply to the material then material undergoes the elastic deformation.

- Yield point is the point where

- material limit the elastic deformation and beginning of the plastic deformation

Point D :- Maximum tensile strength :-

- This point show that material with stand with maximum stress.
- After this point material start with necking although decrease the load, at the point

Point E :- Fracture point :-

- material at the point (E) break called as fracture point.
- Generally material at the point failure occur.

ii) Stress strain diagram of
→ ceramic
polymer
bone
metal.

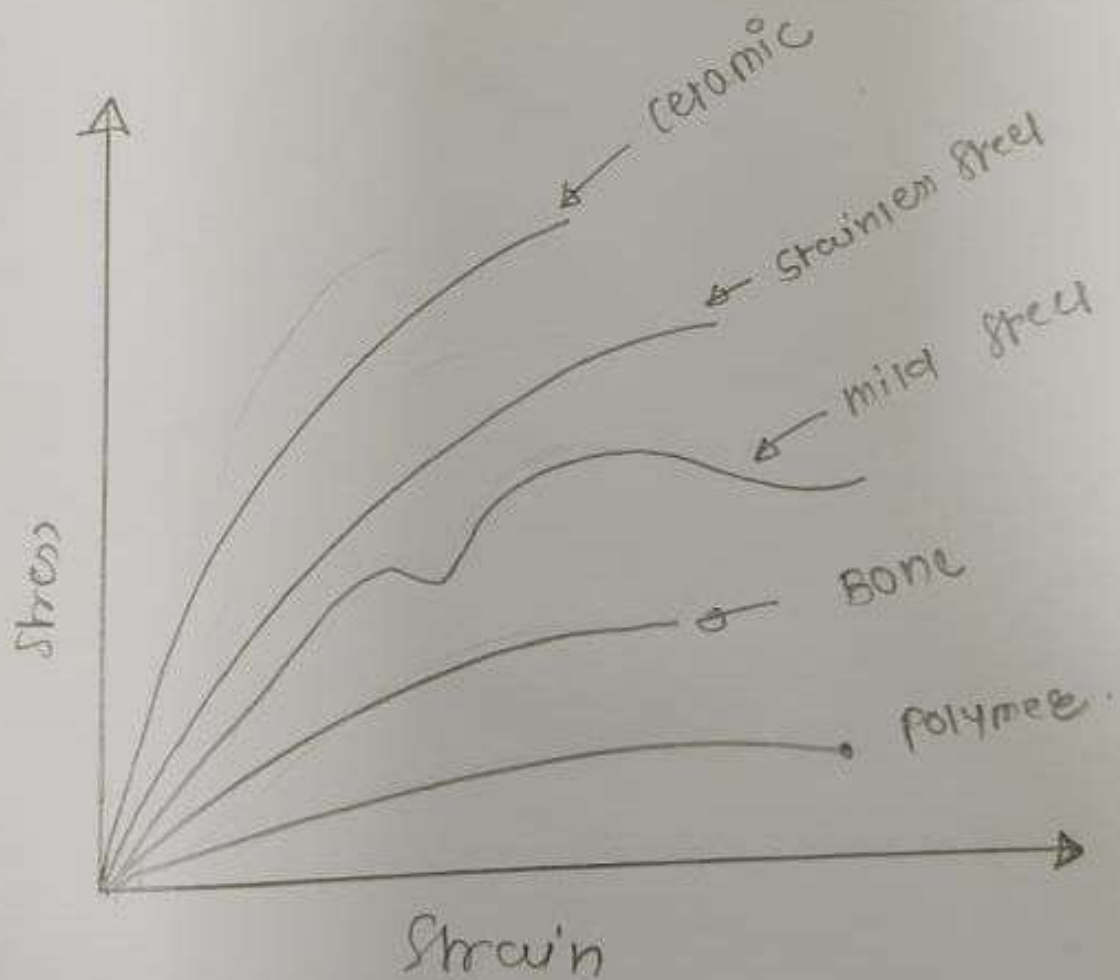


fig 3.

→ This is the stress strain curve diagram of

- ceramic
- polymer
- bone
- metal.